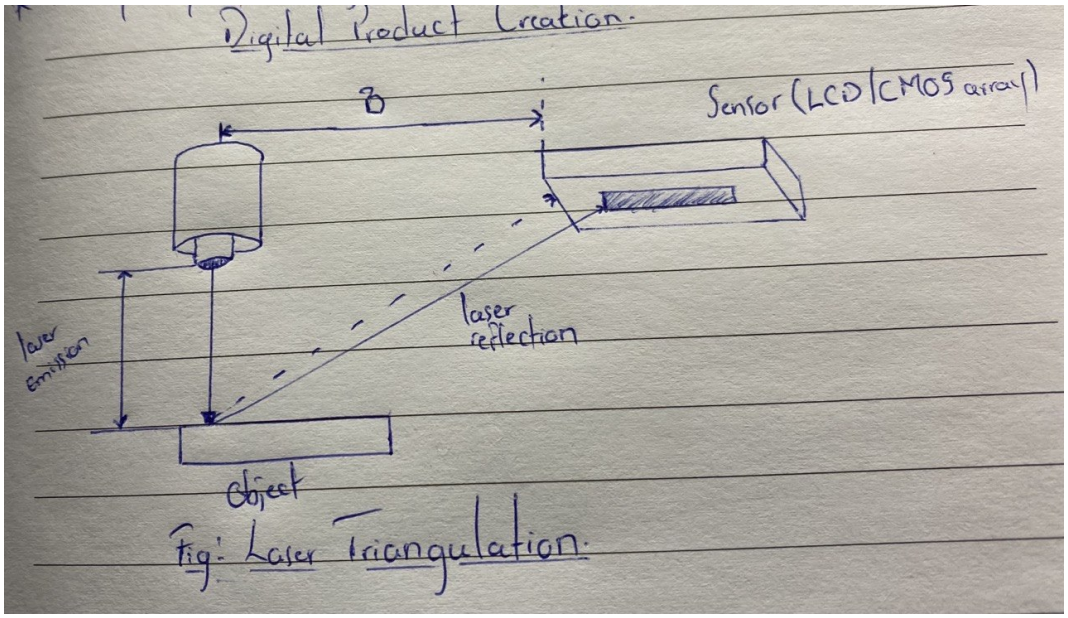
Daily Rant Digital Product Creation

Tadouanla Guetchuin Billy
28177

Today in the center of knowledge **Digital Product Creation**, I had the opportunity to have learn about 3D scanning. **3D scanning** is the process of capturing real-world objects environments to be remodelled or analysed in the digital world. 3D scanning could also be seen as reverse engineering process. For example: in the case where access to a particular part, spare part, or even damage components, 3D scanning could easily be of used in such cases without actually having access to the official dimensions from the manufacturer.

During the lab session we looked into the various scanners and utilizing different scanning technologies and methods. We used a hand scanner, which uses a set of two cameras and laser stripes. In order to scan an object, the object needs to be marked with some special markers as well as the surface. These are used as reference point for the scanner in order for it to locate itself. The marked object is placed on the marked surface and the scanning is done manually by moving the scanner around the object at a particular distance. A scanner capable of automatically scanning objects on a rotating surface was introduced to us. We essentially needed to place the object on a rotating surface and the scanner/projector will scan the object as it rotates on the surface.



$$\frac{X}{f_l} = \frac{B}{Z}$$

$x =$ image coordinate on sensor
$f_l =$ Focal length
$B =$ baseline (Distance Laser-sensor)
$Z =$ Object Distance